

Goniometer

→ Goniometer is an instrument that is used to measure the ROM of any joint of body.

Gonio → Angle

Metron → Measurement.

→ Both active and Passive ROM can be measured.

• Types of Goniometer:-

- Universal Goniometer
- Gravity dependent
- Pendulum goniometer
- Electro goniometer

• Universal goniometer

→ Most widely used

→ Can be used to measure AROM and PROM of almost all joints.

→ May be of Plastic / metal.

→ It has

- Body
- Stable / stationary arm
- Movable arm.

• Body

- Like half / full circle protractor.

- Half circle protractor - 0-180°

- Full " " - 0-360°

→ Axis and fulcrum screw put at middle

→ Axis of goniometer should be placed at joint

• Stable / Stationary arm.

- Extended and fixed arm of goniometer.
- Doesn't have any motion.
- Aligned with prox. segment of jt.

• Movable arm

- Additional attachment with body at axis.
- Aligned with distal segment of jt.
- Moves with joint.

• Gravity dependent goniometer.

- Also k/a fluid goniometer & Pelvic inclinometer
- Have gravity effecting pointer and fluid filled chamber with air bubbles.

- used for measuring Pelvic tilt and Pelvic drop.

• Pendulum goniometer.

- Consists of 360° protractor & weight pointer

Procedure

- Patient's clothes are removed at the point of measure.
- Patient should be in relaxed manner and there shouldn't be any obstruction.
- Joint should be in 0°.
- Process should be explained to patient.
- Therapist - moves the patient facing the jt.

- Axis/fulcrum placed at joint axis.
- Stable arm with proximal segment
- Mov. arm is distal segment.
- Move the mov. arm along the distal part.

Factors affecting joint ROM.

- Soft tissue tightness
 - Muscle tightness
 - Ligament "
 - Capsule "
 - Cartilage "
 - Can be due to
 - Immobilisation.

- Adhesion
- Contracture
- Injury
 - RA
 - AS
 - OA
 - TB arthritis
 - Psoriatic "
 - Bulky muscle
 - Sex
 - Age

Shoulder Jt.

• Flexion

ROM - 0-180°

PP - supine

Axis - Greater tuberosity of humerus

MA - Placed at lateral of arm

SA - Mid axillary line of trunk.

• Extension

ROM - 180-0 / 0-60

PP - Prone

Axis - same

MA - same

SA - same

• Abduction

ROM - ~~0~~ 0-180°

PP - supine

Axis - Ant. of acromion process

SA - slight lat to ant chest ||el to sternum

MA - Ant. of arm ||el to humerus shaft
aligned w $\bar{c}$ medial epi

• Adduction

ROM - 180-0

Everything same as Abel.

• External rotation

ROM - 0-90°

PP - Supine

Sh. abd - 90°

Elbow flexed

Forearm ⊥ and ⊥ to table

Elbow not supported.

Humerus supported

Distal "

Thorax

Scapula

} → Stable

Axis - olecranon process of ulna.

SA - ⊥ to table / ⊥ to floor

MA - Along ulna shaft aligning w/ styloid process of ulna

Elbow jj.

• Flexion / extension.

ROM - 0-145° / 140-0

PP - Supine

Axis - lat. epi

SA - lat midline of humerus

MA - " " " forearm

Radio-ulnar

• Pronation / supi.

ROM -

PP - Sitting

Forearm mid prone

- Axis - At ulnar styloid process
 - Both arm of gonio at 90°

• SA - $\perp$ to mov. arm

• MA - Ant of wrist

Knee

• Flexion

ROM - 0-120

PP - Supine & knee flexed at end motion

Axis - Lat of hip over greater trochanter

MA - Along femur (lat. epi.)

SA - " midline of pelvis

• Extension

ROM - 0-30

PP - Prone

Axis - on greater trochanter

MA - Same as flex.

SA - " " "

• Abduction

ROM - 0-45

PP - Supine

Axis - ASIS of measured limb

MA - Ant midline of femur in the line & midline of patella.

SA - Imaginary horizontal line from one ASIS to other

Adduction

ROM - 0-30 / 45-0

- All same as abd.

Int. Rotation

ROM - 0-45

PP - High sitting, $\bar{c}$ flexed knee

Axis - Patella center.

SA - Vertically

MA - $\bar{c}$ leg

Ext. Rotation

ROM - 0-45

- All same as Int rot